Fiscal Year:	FY 2024	**Task Last Updated:**	FY 08/28/2025
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Project Title:	Cold-Sleep for Long-duration Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	15260	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2021 BRASH2101: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	06/01/2022	**End Date:**	10/31/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2024 per TRISH Science Office (Ed., 3/14/24).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buysse, Daniel M.D. (University of Pittsburgh) Empey, Philip Ph.D. (University of Pittsburgh) Guyette, Francis Ph.D. (University of Pittsburgh) Weissman, Alexandra M.D. (University of Pittsburgh)		
Grant/Contract No.:	NNX16AO69A-T0601		
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Performance Goal Text:			

Task Description:	We propose a protocol for a subset of the crew to adopt a temporary low-resource status, which we call “cold-sleep,” to mitigate psychological stress and reduce resource utilization. Cold-sleep will reduce psychological stress by decreasing perceived duration of mission segments where crewmembers do not have specified tasks. Cold-sleep will reduce oxygen and food consumption as well as carbon dioxide and solid waste production. Cold-sleep uses drugs to increase the normal ratio of sleeping:wakefulness time for a crewmember from ~6 hours:18 hours to ~20 hours:4 hours with a mild 1-2ºC reduction in core body temperature during sleep. This target should reduce perceived mission duration by over 75% and metabolic requirements by 50%. In prior human experiments by our group, we found that the intravenous (IV) alpha-2-adrenergic receptor (A2AR) agonist dexmedetomidine induces a light sleep, mild hypothermia, and reduces resting oxygen consumption (VO2) by 25-30% relative to quiet resting. In a series of laboratory protocols in healthy human volunteers, we will solve several scientific, design, and technical challenges to bring A2AR-drug induced cold-sleep to a complete working prototype. Aim 1: Demonstrate similar metabolic suppression with oral, topical, or transmucosal A2AR agonists. This step is essential to eliminate the need for long-term IV access and to allow self-administration by a crewmember. Aim 2: Demonstrate a comfortable and reliable monitoring array and cooling pad to ensure astronaut safety and temperature control. This step will translate the invasive techniques we find expedient in short-term laboratory experiments to a practical solution that will be palatable and easily used in long-term missions. Aim 3: Demonstrate the actual metabolic savings, perceived time, and rate of recovery to normal function after a short and medium length cold-sleep regimen. This will demonstrate readiness of the cold-sleep protocol. Our protocols will establish the pathway for testing the efficacy of other emerging drugs for metabolic suppression that are not yet approved in humans.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Key Findings: In Aim 1, 31 healthy adult participants completed 54 protocol days. We measured energy expenditure using indirect calorimetry, core body temperature with ingested telemetry capsules, and other physiological variables. Single oral doses of dexmedetomidine (1 µg/kg sublingual or 4 µg/kg swallowed) or tizanidine (8 mg or 16 mg) reduce energy expenditure (13% to 22%) and core body temperature (0.41ºC to 0.64ºC). Decreases in temperature were not associated with increased energy expenditure after high doses of each drug. We conclude that oral dexmedetomidine and tizanidine can reduce energy expenditure and suppress cold-induced thermogenesis in humans. We tested sustained administration of dexmedetomidine (2 µg/kg oral dose followed by 1µg/kg/hr subcutaneous administration for 6 hours) in 11 participants. Temperature decreases 1.5ºC and energy expenditure decreases 20%. Sedating effects of drug lasts 6-8 hours. Decreased core temperature was associated with decreased metabolic rate when plasma drug levels were ~ 1.0 ng/ml. We conclude that ~1.0 ng/ml plasma dexmedetomidine levels create physiological changes that mimic some features of animal hibernation. For Aim 2, we tested swallowed temperature telemetry capsules from (HQI) and (CoreTemp, GreenTeq), an isothermal surface termperature sensor (Spot-On, Bair-Hugger), and a dual-sensor device (Calera Core Temperature Sensor). The HQI system often had loss of connectivity from radio interference. The CoreTemp capsule has an internal memory and rare data loss. The Spot-On device is well correlated with CoreTemp temperature when placed on the forehead (R2=0.79), and Bland-Altman plots reveal small bias (0.13ºC) with acceptable 95% Limits of Agreement (-0.79 to 0.53ºC). The Spot-On device was not reliable when placed on the torso. The Calera sensor placed on the torso is poorly calibrated for measuring core temperatures below about 36.5ºC. We conclude that the Spot-On device placed on the forehead could serve as a non-invasive core temperature monitor in the 34ºC-37ºC range. Surface temperature and heat flux monitors on the forehead, arm, leg and torso of 21 participants during 4-hour protocols confirms that skin temperature at the forehead was best correlated with core temperature with a mean bias of -2.48 °C and 95% Limits of Agreement of [-2.60ºC to -2.37ºC]. We calculated net heat balance as the difference between energy production and heat loss from the skin. Net heat balance predicted change in temperature. Cooling occurred with a heat loss of 125 W with 60 W provided by a device, when heat production was suppressed with drugs from 81 W to 69 W. Cooling pads circulated water at 20ºC to 25ºC. In non-drugged participants energy production rose to 90 W, net heat balance did not change and cooling did not occur. We conclude that effective temperature control requires both effective heat removal and metabolic control with drugs. In Aim 3, 19 volunteers screened and 12 participated in a 1-day study where they received 2 µg/kg oral dexmedetomidine followed by an 18 hour subcutaneous 0.5 µg/kg/hr (n=6) or 1.0 µg/kg/hr (n=6) infusion. Plasma drug levels were ~0.75 ng/ml or ~1.2 ng/ml respectively. Metabolic rate declined ~20% from baseline in participants, and core body temperature decreased 1.0 - 1.5ºC. Participants were able to complete psychophysiological tests and an exercise regimen at 2-6 hours after stopping infusions. Reaction time increased at 2 hours but recovered to baseline by 6 hours after infusion. We conclude that dexmedetomidine is tolerated for 18 hours by the majority of participants, reduces metabolic rate during rest, and prolongs quiet time to 18-20 hours per day. Of 12 volunteers, 5 participated in a 5-day study of 2 µg/kg oral dexmedetomidine followed by 18 hour subcutaneous infusion of 1.0 µg/kg/hr dexmedetomidine, repeated for 5 days. Peak plasma drug levels were 1.0 - 1.2 ng/ml. On the first two days, metabolic rate declined ~20% from baseline, and core body temperature decreased 1.5ºC-2.0ºC. Drug effects decreased on days 3-5, with sedation lasting only 10-12 hours by the fifth day, and metabolic rate decrease only ~10%. Three participants reported rebound insomnia and anxiety for 24-48 hours after protocol. No clinically significant weight loss, change in body composition, bone composition, or serum chemistries occurred over 5 days.

	We conclude that this drug regimen produces tolerance and becomes ineffective for metabolic suppression after 2 days.
Bibliography Type:	Description: (Last Updated: 11/14/2025)
Articles in Peer-reviewed Journals	Callaway CW, Flickinger KL, Weissman A, Guyette FX, DeMaio R, Jonsson A, Wu V, Monteleone JL, Prescott P, Birabaharan J, Buysse DJ, Empey PE, Nolin TD, West III RE. "Alpha-2-adrenergic agonists reduce resting energy expenditure in humans during external cooling." Temperature. 2024 Apr 21;11(3):280-98. https://doi.org/10.1080/23328940.2024.2339781 ; PMID: 39193049; PMCID: PMC11346546 , Apr-2024
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Awards	Flickinger K. "B-Sure Fellowship, June 2023." Jun-2023